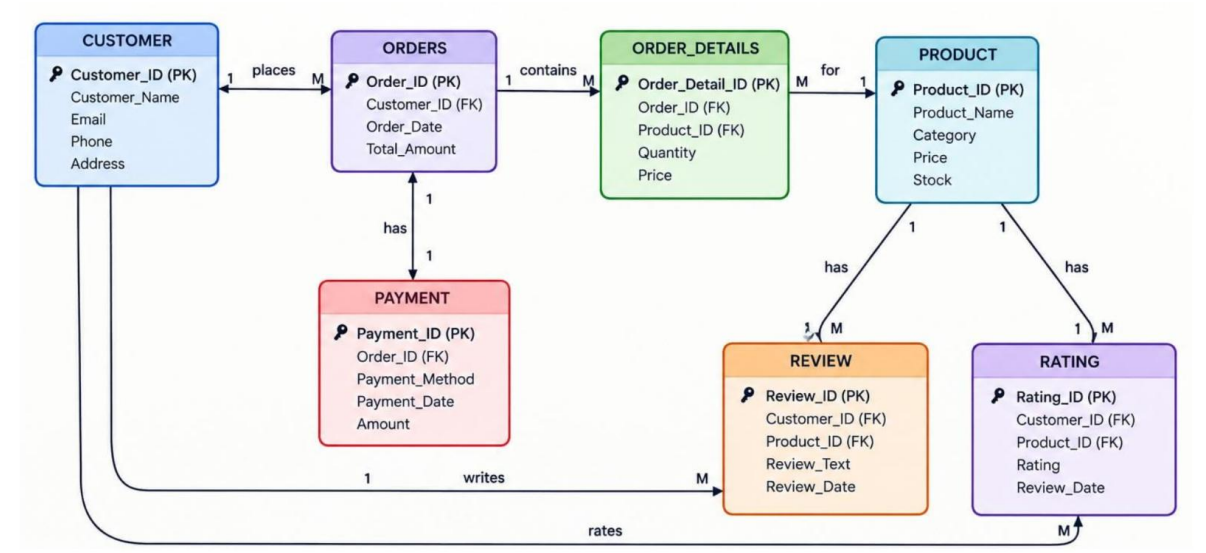
Gaming-Accessories-E-Commerce-Management-System

DATABASE RELATIONSHIP ANALYSIS USING JOINS

Chapter 9

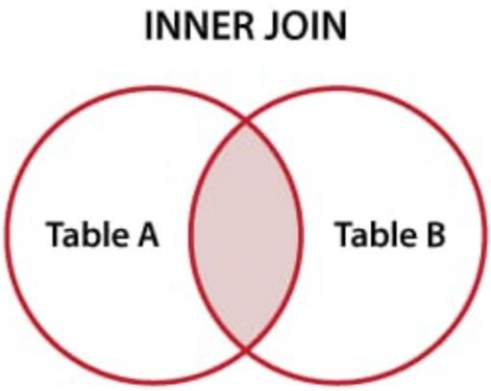
1.Combining Multiple Tables

Combining multiple tables means retrieving related data from two or more tables using SQL JOIN operations and by combining customer, order, product, and payment details.



2.INNER JOIN

The INNER JOIN selects and returns only the rows that have matching values in both tables.



2.1 Customer and Orders

```
SQL> SELECT
  2      C.Customer_ID,
  3      C.First_Name,
  4      C.Last_Name,
  5      O.Order_ID,
  6      O.Order_Date,
  7      O.Total_Amount
  8  FROM Customer C
  9  INNER JOIN Orders O
 10      ON C.Customer_ID = O.Customer_ID;

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
TOTAL_AMOUNT
------------
        101 Rahul
Sharma                                                401 01-JUL-26
        2999

        102 Priya
Kumar                                                 402 03-JUL-26
        1999

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
TOTAL_AMOUNT
------------
```

2.2 Orders and payment

```
SQL> SELECT
  2      O.ORDER_ID,
  3      P.PAYMENT_ID,
  4      P.PAYMENT_MODE,
  5      P.PAYMENT_DATE,
  6      P.PAYMENT_AMOUNT,
  7      P.PAYMENT_STATUS
  8  FROM ORDERS O
  9  INNER JOIN PAYMENT P
 10      ON O.ORDER_ID = P.ORDER_ID;

  ORDER_ID PAYMENT_ID PAYMENT_MODE         PAYMENT_D PAYMENT_AMOUNT
---------- ---------- -------------------- --------- --------------
PAYMENT_STATUS
--------------------
       101        501 UPI                  29-SEP-26           1898
Successful

       102        502 Credit Card          29-SEP-26           1299
Successful
```

2.3 Product and Order details

```
SQL> SELECT
  2      C.Customer_ID,
  3      C.First_Name,
  4      C.Last_Name,
  5      O.Order_ID,
  6      O.Order_Date,
  7      O.Total_Amount
  8  FROM Customer C
  9  INNER JOIN Orders O
 10      ON C.Customer_ID = O.Customer_ID;

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
TOTAL_AMOUNT
------------
        101 Rahul
Sharma                                                401 01-JUL-26
        2999

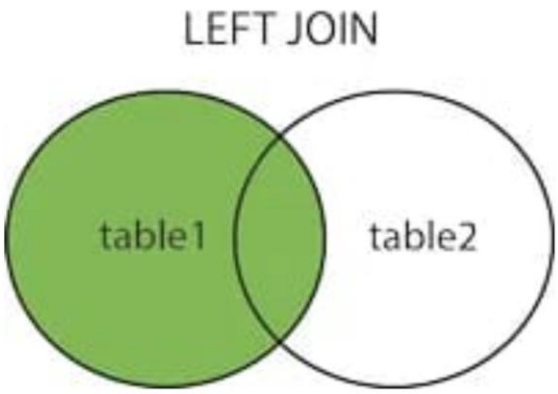
        102 Priya
Kumar                                                 402 03-JUL-26
        1999

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
TOTAL_AMOUNT
------------

        103 Arun
Raj                                                   403 05-JUL-26
        4999
```

3.LEFT JOIN

LEFT JOIN displays all records from the left table and matching records from the right table.



3.1 Display all customers and their orders

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.CUSTOMER_NAME,
  4      O.ORDER_ID,
  5      O.ORDER_DATE,
  6      O.TOTAL_AMOUNT
  7  FROM CUSTOMERS C
  8  RIGHT JOIN ORDERS O
  9      ON C.CUSTOMER_ID = O.CUSTOMER_ID;

CUSTOMER_ID
-----------
CUSTOMER_NAME
----------------------------------------
  ORDER_ID ORDER_DAT TOTAL_AMOUNT
---------- --------- ------------


       401 01-JUL-26         2999



       402 03-JUL-26         1999

CUSTOMER_ID
-----------
CUSTOMER_NAME
----------------------------------------
  ORDER_ID ORDER_DAT TOTAL_AMOUNT
---------- --------- ------------



       403 05-JUL-26         4999
```

4.RIGHT JOIN

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.CUSTOMER_NAME,
  4      O.ORDER_ID,
  5      O.ORDER_DATE,
  6      O.TOTAL_AMOUNT
  7  FROM CUSTOMERS C
  8  RIGHT JOIN ORDERS O
  9      ON C.CUSTOMER_ID = O.CUSTOMER_ID;

CUSTOMER_ID
-----------
CUSTOMER_NAME
----------------------------------------
  ORDER_ID ORDER_DAT TOTAL_AMOUNT
---------- --------- ------------


       401 01-JUL-26         2999



       402 03-JUL-26         1999

CUSTOMER_ID
-----------
CUSTOMER_NAME
```

4.1 Display all orders and customer details

```
SQL> SELECT
  2      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  3      O.ORDER_ID,
  4      O.ORDER_DATE,
  5      P.PRODUCT_NAME,
  6      OD.QUANTITY,
  7      OD.UNIT_PRICE,
  8      PY.PAYMENT_MODE,
  9      PY.PAYMENT_STATUS
 10  FROM CUSTOMER C
 11  INNER JOIN ORDERS O
 12      ON C.CUSTOMER_ID = O.CUSTOMER_ID
 13  INNER JOIN ORDER_DETAILS OD
 14      ON O.ORDER_ID = OD.ORDER_ID
 15  INNER JOIN PRODUCT P
 16      ON OD.PRODUCT_ID = P.PRODUCT_ID
 17  LEFT JOIN PAYMENT PY
 18      ON O.ORDER_ID = PY.ORDER_ID;

CUSTOMER_NAME
--------------------------------------------------------------------
  ORDER_ID ORDER_DAT
---------- ---------
PRODUCT_NAME
--------------------------------------------------------------------
  QUANTITY UNIT_PRICE PAYMENT_MODE         PAYMENT_STATUS
---------- ---------- -------------------- --------------------
Arun Raj
       403 05-JUL-26
Gaming Keyboard
         1       2799


CUSTOMER_NAME
--------------------------------------------------------------------
  ORDER_ID ORDER_DAT
---------- ---------
PRODUCT_NAME
--------------------------------------------------------------------
  QUANTITY UNIT_PRICE PAYMENT_MODE         PAYMENT_STATUS
```

5. Complete Order details

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4      O.ORDER_ID,
  5      O.ORDER_DATE,
  6      O.ORDER_STATUS,
  7      P.PRODUCT_ID,
  8      P.PRODUCT_NAME,
  9      OD.QUANTITY,
 10      OD.UNIT_PRICE,
 11      OD.SUBTOTAL,
 12      O.TOTAL_AMOUNT,
 13      PY.PAYMENT_MODE,
 14      PY.PAYMENT_STATUS
 15  FROM CUSTOMER C
 16  INNER JOIN ORDERS O
 17      ON C.CUSTOMER_ID = O.CUSTOMER_ID
 18  INNER JOIN ORDER_DETAILS OD
 19      ON O.ORDER_ID = OD.ORDER_ID
 20  INNER JOIN PRODUCT P
 21      ON OD.PRODUCT_ID = P.PRODUCT_ID
 22  LEFT JOIN PAYMENT PY
 23      ON O.ORDER_ID = PY.ORDER_ID
 24  ORDER BY O.ORDER_ID;

CUSTOMER_ID
-----------
CUSTOMER_NAME
----------------------------------------------------------------
  ORDER_ID ORDER_DAT ORDER_STATUS                   PRODUCT_ID
---------- --------- ------------------------------ ----------
PRODUCT_NAME
----------------------------------------------------------------
  QUANTITY UNIT_PRICE   SUBTOTAL TOTAL_AMOUNT PAYMENT_MODE
---------- ---------- ---------- ------------ ----------------
PAYMENT_STATUS
--------------------
        101

CUSTOMER_ID
```

6.Customer purchase history

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4      O.ORDER_ID,
  5      O.ORDER_DATE,
  6      O.ORDER_STATUS,
  7      P.PRODUCT_NAME,
  8      OD.QUANTITY,
  9      OD.UNIT_PRICE,
 10      OD.SUBTOTAL,
 11      O.TOTAL_AMOUNT
 12  FROM CUSTOMER C
 13  INNER JOIN ORDERS O
 14      ON C.CUSTOMER_ID = O.CUSTOMER_ID
 15  INNER JOIN ORDER_DETAILS OD
 16      ON O.ORDER_ID = OD.ORDER_ID
 17  INNER JOIN PRODUCT P
 18      ON OD.PRODUCT_ID = P.PRODUCT_ID
 19  ORDER BY C.CUSTOMER_ID, O.ORDER_DATE;

CUSTOMER_ID
-----------
CUSTOMER_NAME
----------------------------------------------------------------
  ORDER_ID ORDER_DAT ORDER_STATUS
---------- --------- ------------------------------
PRODUCT_NAME
----------------------------------------------------------------
  QUANTITY UNIT_PRICE   SUBTOTAL TOTAL_AMOUNT
---------- ---------- ---------- ------------
        101
Rahul Sharma
       401 01-JUL-26 Delivered

CUSTOMER_ID
-----------
CUSTOMER_NAME
----------------------------------------------------------------
  ORDER_ID ORDER_DAT ORDER_STATUS
```

7.Customer purchase history with payment

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  4      O.ORDER_ID,
  5      O.ORDER_DATE,
  6      O.ORDER_STATUS,
  7      P.PRODUCT_NAME,
  8      OD.QUANTITY,
  9      OD.UNIT_PRICE,
 10      OD.SUBTOTAL,
 11      O.TOTAL_AMOUNT,
 12      PY.PAYMENT_MODE,
 13      PY.PAYMENT_AMOUNT,
 14      PY.PAYMENT_STATUS
 15  FROM CUSTOMER C
 16  INNER JOIN ORDERS O
 17      ON C.CUSTOMER_ID = O.CUSTOMER_ID
 18  INNER JOIN ORDER_DETAILS OD
 19      ON O.ORDER_ID = OD.ORDER_ID
 20  INNER JOIN PRODUCT P
 21      ON OD.PRODUCT_ID = P.PRODUCT_ID
 22  LEFT JOIN PAYMENT PY
 23      ON O.ORDER_ID = PY.ORDER_ID
 24  ORDER BY C.CUSTOMER_ID, O.ORDER_DATE;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------
  ORDER_ID ORDER_DAT ORDER_STATUS
---------- --------- ------------------------------
PRODUCT_NAME
--------------------------------------------------------------
  QUANTITY UNIT_PRICE   SUBTOTAL TOTAL_AMOUNT PAYMENT_MODE
---------- ---------- ---------- ------------ ----------------
PAYMENT_AMOUNT PAYMENT_STATUS
-------------- --------------------
       101
```

8.Multi-Table business report

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME
  4      COUNT(DISTINCT O.ORDER_ID) AS TOTAL_ORDERS,
  5      SUM(OD.SUBTOTAL) AS TOTAL_PURCHASE,
  6      SUM(OD.QUANTITY) AS TOTAL_PRODUCTS
  7  FROM CUSTOMER C
  8  INNER JOIN ORDERS O
  9      ON C.CUSTOMER_ID = O.CUSTOMER_ID
 10  INNER JOIN ORDER_DETAILS OD
 11      ON O.ORDER_ID = OD.ORDER_ID
 12  GROUP BY
 13      C.CUSTOMER_ID,
 14      C.FIRST_NAME,
 15      C.LAST_NAME
 16  ORDER BY TOTAL_PURCHASE DESC;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------
TOTAL_ORDERS TOTAL_PURCHASE TOTAL_PRODUCTS
------------ -------------- --------------
        104
Sneha Reddy
           1           5998              2

        101
Rahul Sharma
           1           5997              3
```

9.Payment transaction report

```
SQL> SELECT
  2      PY.PAYMENT_ID,
  3      O.ORDER_ID,
  4      C.CUSTOMER_ID,
  5      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  6      PY.PAYMENT_MODE,
  7      PY.PAYMENT_DATE,
  8      PY.PAYMENT_AMOUNT,
  9      PY.PAYMENT_STATUS,
 10      O.TOTAL_AMOUNT,
 11      O.ORDER_STATUS
 12  FROM PAYMENT PY
 13  INNER JOIN ORDERS O
 14      ON PY.ORDER_ID = O.ORDER_ID
 15  INNER JOIN CUSTOMER C
 16      ON O.CUSTOMER_ID = C.CUSTOMER_ID
 17  ORDER BY PY.PAYMENT_DATE DESC;

PAYMENT_ID   ORDER_ID CUSTOMER_ID
---------- ---------- -----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
PAYMENT_MODE         PAYMENT_D PAYMENT_AMOUNT PAYMENT_STATUS       TOTAL_AMOUNT
-------------------- --------- -------------- -------------------- ------------
ORDER_STATUS
------------------------------
       501        401         101
Rahul Sharma
UPI                  29-OCT-26           2999 Completed                    2999
Delivered


PAYMENT_ID   ORDER_ID CUSTOMER_ID
---------- ---------- -----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
PAYMENT_MODE         PAYMENT_D PAYMENT_AMOUNT PAYMENT_STATUS       TOTAL_AMOUNT
-------------------- --------- -------------- -------------------- ------------
ORDER_STATUS
------------------------------
       502        402         102
Priya Kumar
Card                 29-OCT-26           1999 Completed                    1999
Shipped
```